Engineering Standard

Fastener selection and corrosion protection for chassis-mounted components

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Owner	Chassis Systems Standards Committee
Approved by	Chief Engineer - Chassis Systems
Applies to	All chassis-mounted brackets, mountings and fixings, HCV platforms
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Review due	August 2026

Summary

This standard defines the fastener families, joint types and coating systems permitted on chassis-mounted components, and the salt-spray performance each application must meet. It is normative: where a design departs from it, a deviation must be raised and approved. Rev 04 reduces the permitted fastener population from eighty-four part numbers to thirty-one and withdraws 8 um clear zinc plating from all exposed chassis applications.

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1 Scope and normative language

This standard applies to all threaded fasteners and their associated coatings on chassis-mounted components across the HCV platforms. It does not apply to powertrain internal fasteners, to fasteners within supplier-sealed assemblies, or to body-builder interfaces beyond the chassis interface plate.

1.1 Normative language

- Shall indicates a requirement. Departure requires an approved deviation.
- Should indicates a recommendation. Departure requires a recorded engineering justification but not a deviation.
- May indicates a permitted option carrying no preference.

1.2 Relationship to value-engineering proposals

A value-engineering proposal that changes a fastener, a coating or a joint type shall state its position against this standard. A proposal that would require a deviation shall say so at submission. Proposals that reduce cost by moving to a coating below the requirement in section 3 shall not be accepted.

> Note. Eleven proposals in FY24 were closed because they proposed a coating system that does not meet the chassis requirement. In nine of those the originator was unaware of this standard. Originators should check section 3 before preparing a coating-change case.

2 Permitted fastener families

2.1 Rationalised family list

Rev 04 reduces the permitted population to thirty-one part numbers across five families. New designs shall select from this list. Existing part numbers outside the list remain valid for service but shall not be specified on new designs.

Family	Thread sizes	Grade	Drive	Typical application
F1 - Flanged bolt, serrated	M8, M10, M12, M16	10.9	Hex flange	Bracket to frame, primary
F2 - Flanged bolt, plain face	M8, M10, M12	8.8	Hex flange	Bracket to bracket
F3 - Hex bolt with washer	M10, M12, M16, M20	10.9	Hex	Where flange will not seat
F4 - Stud and flanged nut	M12, M16, M20	10.9	-	Spring and axle fixings
F5 - Shoulder bolt	M10, M12	12.9	Hex socket	Pivot and bush fixings

2.2 Selection rules

- Primary structural joints between a bracket and the frame shall use family F1.
- A plain washer shall not be used under a flanged fastener.
- Where a prevailing-torque feature is required, it shall be achieved by the serrated flange of family F1 and shall not be achieved by a separate lock washer.
- Grade 12.9 shall not be used in applications exposed to hydrogen-charging coating processes.

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- A joint shall not mix a metric fastener with a thread form of different pitch within the same bracket.

Requirement 2.1. Separate lock washers are withdrawn. Analysis of 188 fastener-loosening claims in FY24 found the plain washer under flanged nut arrangement in 71 per cent of cases. See CVOEM/VAVE/WF-FA/2024/021 section 5.1.

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3 Corrosion protection requirements

3.1 Requirement by exposure class

Exposure class	Definition	Requirement	Permitted systems
C1 - Exposed chassis	Directly exposed to road spray	720 h to red rust	Zinc-nickel 8 um; zinc flake; E-coat plus topcoat
C2 - Sheltered chassis	Shielded from direct spray	480 h to red rust	Zinc flake; E-coat; zinc-nickel
C3 - Cab interior	Within sealed cab volume	168 h to red rust	Zinc plating 12 um yellow; E-coat
C4 - Enclosed assembly	Within a sealed housing	96 h to red rust	Any system in section 3.2

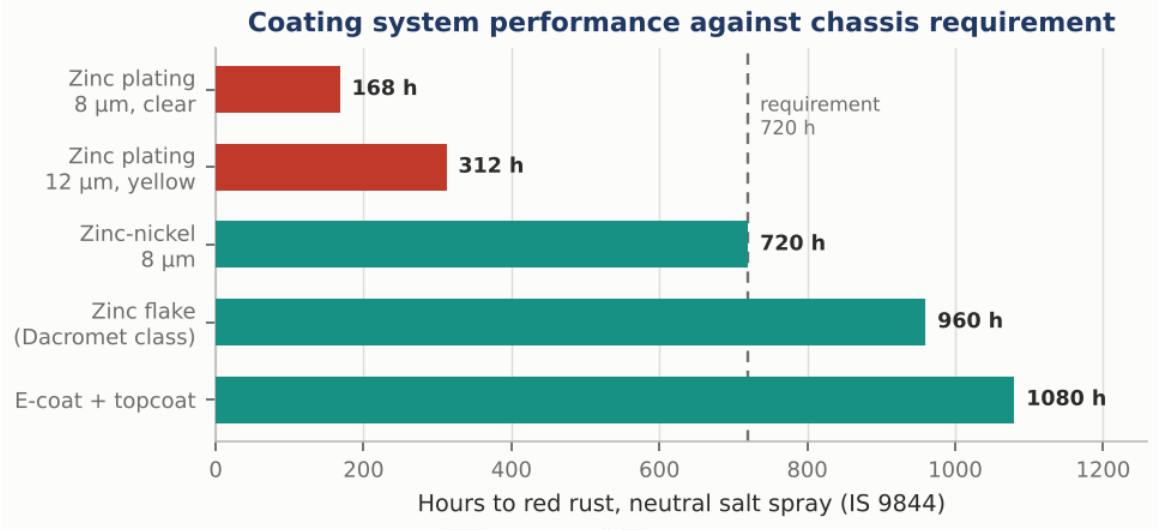


Figure 1 - Coating system performance against the C1 chassis requirement of 720 hours to red rust. Systems below the requirement are withdrawn from exposed chassis applications.

Requirement 3.1. Zinc plating at 8 um clear and 12 um yellow shall not be specified for exposure class C1. Both fall below the 720 hour requirement, at 168 and 312 hours respectively. Existing C1 part numbers using these systems were transitioned during FY24.

3.2 Test method

Performance shall be demonstrated by neutral salt spray to IS 9844, with red rust on the significant surface as the end point. White rust is not an end point for these classes. Testing shall be on production-representative parts including any forming operations performed after coating, since forming after coating materially reduces performance on zinc flake systems.

4 Joint types and locking features

4.1 Permitted joint types

Joint type	Description	Locking feature	Where permitted
J1	Flanged bolt into tapped hole	Serrated flange	C1 to C4
J2	Flanged bolt, nut and plain hole	Serrated flange both ends	C1 to C4
J3	Stud with flanged nut	Serrated flange	C1, C2
J4	Bolt with structural washer	Prevailing torque nut	Where flange cannot seat
J5	Shoulder bolt in bush	Thread adhesive	C3, C4 only

4.2 Dissimilar material joints

Where a joint clamps components of materially different hardness, the design shall address fretting at the interface. Acceptable measures are a hardened steel insert, a bonded interface, or a surface treatment demonstrated to resist fretting at the expected relative displacement.

Requirement 4.1. A clamped joint between aluminium and steel shall be validated on a fixture that reproduces the joint, not on a rigid interface plate. A component-level fatigue result obtained without the joint shall not be accepted as evidence for this class of design. See CVOEM/VAVE/WF-FA/2024/021 finding 3.2.

4.3 Torque specification

Size	Grade 8.8 (Nm)	Grade 10.9 (Nm)	Grade 12.9 (Nm)
M8	24	34	40
M10	47	67	79
M12	82	115	135
M16	200	285	335
M20	390	555	650

Values are for a zinc-nickel coated fastener with a friction coefficient of 0.12 to 0.18. Other coatings require recalculation.

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5 Deviations

A deviation shall be raised where a design departs from any requirement in this standard. The deviation request shall state the requirement departed from, the engineering justification, the compensating measure, and the validation evidence. Deviations are approved by the Chassis Systems Standards Committee and are recorded against the part number, not against the project.

Deviation class	Approval authority	Validity
D1 - Coating below class requirement	Chief Engineer	Single part number, 24 months
D2 - Fastener outside permitted family	Standards Committee	Single part number, 36 months
D3 - Joint type not listed	Standards Committee	Single application
D4 - Torque outside table	Standards Committee	Single part number

6 Change record

Rev	Date	Change
01	14 Mar 2019	First issue
02	22 Jul 2021	Added exposure class C4; revised torque table
03	11 Feb 2023	Added zinc-nickel; withdrew cadmium systems
04	30 Aug 2024	Fastener family rationalisation from 84 to 31 part numbers; withdrew 8 um clear and 12 um yellow zinc from class C1; added section 4.2 on dissimilar material joints; withdrew separate lock washers

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